MOCK - Choosing a Stack or Queue
SYNTHETIC LOCAL DEMO - NOT INSTITUTIONAL CONTENT OR APPROVAL

Objective: Apply last-in-first-out and first-in-first-out rules to a short sequence of stored items.

MOCK ORIGINAL LESSON. A stack removes the most recently added item first: last in, first out (LIFO). A queue removes the earliest waiting item first: first in, first out (FIFO). The same stored items can therefore leave in different orders.

WORKED EXAMPLE
Add A, then B, then C. In a stack, the next removal returns C because it was added last. In a queue, the next removal returns A because it has waited longest. Both structures then contain two items; their removal rule, rather than their item count, distinguishes them.

CHECK AND CORRECTION
An undo history should return the most recent action first. Which structure matches that rule?
A stack returns the latest stored action first. This is a rule-selection example, not a full undo implementation.

Keep order and count separate. Enqueue adds one item at the back; dequeue removes one item from the front. After four additions and two valid removals from a previously empty queue, two items remain. A removal from an empty structure is outside this example and must be handled explicitly in real software.

INDEPENDENT APPLICATION
An empty queue receives four items, then performs two valid removals. How many items remain?
Four additions produce four items. Two removals leave 4 - 2 = 2 items.

GLOSSARY
Stack: A structure whose next removal returns the most recently added remaining item.
Queue: A structure whose next removal returns the earliest added remaining item.
FIFO: First in, first out: the earliest waiting item leaves first.
LIFO: Last in, first out: the most recently added item leaves first.

Choose the structure from the required removal rule, then trace its operations. Stack means latest first; queue means earliest first. State empty-structure behavior separately rather than guessing.

Original mock fixture. No real learner data, school policy, signatures or redistribution review.
Version 1. Accessible lesson text and authored checks are included separately in this package.